



Optimal participation of battery swapping stations in frequency regulation market considering uncertainty

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ARTICLE INFO

Handling editor: Henrik Lund

Keywords:

Electric vehicle
Battery swapping
Frequency regulation
Real-time response
Reserve capacity
Information-gap decision theory

ABSTRACT

Electric vehicle battery swapping stations (BSS) have significant potential in power system frequency regulation. However, uncertainties of swapping demand and regulation signals introduce risks to operational benefits and regulation performances. Aiming to enhance overall profitability, this study proposes day-ahead bidding and real-time scheduling strategies for BSS to participate in frequency regulation. In the day-ahead stage, the BSS scheduling model is established. Then, the information-gap decision theory (IGDT) is employed to determine the optimal reserve capacities. The randomness of regulation signals is captured by designing a new uncertainty set based on statistical analysis. Additionally, a boundary autonomous selection (BAS) method is developed to solve the IGDT model, making the selected scenarios more consistent with real-world situations. In the intra-day stage, a real-time response strategy is proposed to track the regulation signals considering the BSS operation economy. Simulations are conducted using historical data from Pennsylvania, Jersey, and Maryland (PJM). The results demonstrate that the reserve capacities and response ratios obtained by the proposed strategies yield higher profits for BSS while ensuring regulation performance.

Nomenclature

Indexes and sets	
τ	Index of the time when the regulation signal is sent
t	Index of scheduling hour
i	Index of inventory battery
k	Index of different day
Ω_t	Set of the time in hour t
Parameters	
κ	Regulation signal at τ
$\Delta\tau$	Time interval between adjacent regulation signals
$P^{ch/dis,max}$	Rated charging/discharging power of BSS charger
$N^{ch/ba}$	Number of chargers and inventory batteries in BSS
$\eta^{ch/dis}$	Charging/discharging efficiency
$\bar{E}/\underline{E}$	Energy of fully-charged/depleted battery
$\tilde{N}^{sw}$	Predicted swapping demand
$\tilde{\pi}$	Mean HCRS
N^{sf}	Number of swapping facilities
$\lambda^{sw/\pi}$	Weights of uncertainty radius
$\lambda^{rc/res/sw/que/grid/pun}$	Reserve capacity price/regulation response price/swapping price/coefficient of queueing penalty/energy price/coefficient of deployment shortage punishment
Q	Number of binary variables in the quasi-precision method

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(continued)

$\underline{\gamma}$	Lower bound of response ratio for a single regulation signal
$\underline{\gamma}^{ave}$	Lower bound of average response ratio
Variables	
π	HCRS
$\pi^{\pm}$	Positive/negative HCRS
R	Regulation reserve capacity
P^{da}	Baseline power
D	Power variation caused by frequency regulation
$D^{\pm}$	Power variation caused by up/down regulation
P/P_i	Power of BSS/battery i
$P^{ch/dis}$	Charging/discharging power
E^{bss}/E_i	Energy of BSS/battery i
x	Binary variable that determines whether the battery is swapped
QUE	Number of queued EVs
$P^{reg/sw/en/pun}$	Regulation revenue/swapping revenue/energy cost/deployment shortage punishment
N^{sw}	Swapping demand
α	Uncertainty radius
$\alpha^{sw/\pi}$	Uncertainty radius of swapping demand/HCRS
$\bar{\pi}/\underline{\pi}$	Maximum up/down fluctuation of HCRS
γ	Response ratio to the regulation signal
γ^{ave}	Average response ratio of j signals
μ	Relaxation variable

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<https://doi.org/10.1016/j.energy.2024.131815>

Received 8 October 2023; Received in revised form 23 May 2024; Accepted 25 May 2024

Available online 27 May 2024

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1. Introduction

1.1. Motivation for this work

To address energy and environmental challenges, the large-scale integration of renewable energy sources has become a development trend of the power system. In some distribution networks, distributed generation (DG) capacity even exceeds the total load. The strong intermittency and variability of DG make it increasingly difficult to balance the supply and demand of the grid [1]. The primary and secondary regulation of the traditional generators cannot completely suppress the resulting frequency fluctuation. Therefore, providing ancillary frequency regulation through demand-side flexible loads has become a new solution [2]. As a typical flexible load, the EV charger is ideal for mitigating short-term frequency fluctuations for its fast ramp rates and significant demand response potential [3]. However, plug-in EV charging comes with strong subjective randomness. As EVs sparsely connect to the grid, ensuring an adequate regulation reserve capacity remains challenging. In recent years, a new EV battery swapping model has flourished. EVs can replace a fully-charged battery at a battery swapping station (BSS) within a few minutes. EV manufacturer NIO, for example, has established more than 2300 BSSs in China by March 2024 [4]. BSS owns the battery assets and manages the charging control centrally. The large inventory of batteries resembles a battery energy storage system (BESS), making it easier for BSS to meet the reserve capacity requirement of 1 MW [5]. Therefore, BSS has a natural advantage over plug-in EVs regarding frequency regulation capabilities. The strategy of BSS participating in the frequency regulation market needs to be tackled.

1.2. Background and research gaps

The business model of battery swapping includes many participants such as BSS operators, EV owners and utilities. Under the traditional tariff-based model, BSS can already use its BESS-similar properties to interact with the grid [6]. As the model has limitations in mobilizing the participation of many stakeholders, the participants under the battery swapping framework are also exploring new business models. GLP has partnered with CATL to operate BSS with the Energy-as-a-service (EaaS) solution to serve heavy trucks, expanding the use of clean energy and promoting the sustainable development of the transportation industry [7]. Some studies have focused on the importance of EVs in the business model by proposing new paradigms such as Mobile-Energy-as-a-Service (MEaaS) [8], or by developing a battery swap pricing model [9] to promote traffic and grid flexibility. BSS's low-power charging management of batteries can effectively slow down battery degradation, giving birth to the Battery-as-a-Service (BaaS) business model [10]. Companies such as NIO are committed to promoting this paradigm, where battery swapping companies lease batteries to EV owners rather than sell them. It can be seen that the business models under the battery swapping framework are diversified, but the purpose is to improve the profitability or operational performance of all participants, which is consistent with the aim of this paper.

In the United States, the Federal Energy Regulatory Commission (FERC) issued Order 755 to provide policy guidance for the participation of demand-side resources in frequency regulation markets [11]. Today, operators like Pennsylvania, Jersey, and Maryland (PJM) and the California Independent System Operator (CAISO) have established sophisticated day-ahead bidding and intra-day regulation mechanisms for these resources (known as dynamic resources). In China, grid companies in Guangdong and Inner Mongolia have also established frequency regulation markets.

The market share of electric vehicles (EVs) is rapidly increasing today. According to the latest forecasts from the International Energy

Agency (IEA), global EV battery demand is expected to reach 3500 GWh by 2030 [12]. Thanks to the wide application of high-power chargers and vehicle-to-grid (V2G) technology, the frequency regulation capability of EVs has received extensive attention [13].

Many articles have investigated the ability of EV chargers to provide frequency regulation. Most studies use EV aggregators or signing contracts with EV owners to ensure the available reserve capacity [14]. In [15], a simplified modeling approach for EV aggregators providing frequency regulation is proposed using the state-space method. A hierarchically adaptive strategy is developed in [16] to control EVs and power plants at different levels for frequency regulation. Based on constructing the model, some researchers pay attention to the economic interests, which directly affect the willingness of EVs to participate. Among them [17], validates from an economic perspective that EVs are able to provide frequency regulation services [18,19]. maximize regulation benefits by proposing regulation price prediction models. The price is further taken as a variable in [20] and optimized by employing the generalized Stackelberg game to incentivize the proactive participation of EVs in the regulation market. Frequency regulation inevitably cause fluctuations in EVs' state-of-charge (SOC). Based on introducing a SOC-based judgment module by [21], whether EVs participate in regulation service can be automatically assessed. At a faster time-step, an optimal control method that can simultaneously address the EVs' SOC requirements and the aggregator's reserve revenue is proposed in [22]. [23] further allocates the regulation task within the frequency regulation capacity of EVs according to their SOC. While numerous studies have validated the flexible control of EVs in frequency regulation, there is still a lack of research on the optimality of regulation reserves.

Frequency regulation resources must bid for hourly reserve capacities in day-ahead ancillary markets. Then, during intra-day operation, they provide ancillary frequency regulation by real-time power control. To determine the reserve capacities [24], performs day-ahead scheduling by carefully considering the EV's SOC requirement and battery degradation [25,26]. utilize game theory in the decision-making process for reserve capacity to trade-off the interests of various stakeholders. Furthermore, it's challenging for regulation resources to fully respond to the regulation requirements of the independent system operator (ISO). The model predictive control (MPC) is a real-time scheduling method commonly used in regulation control [27,28]. [29] further employs an economic MPC (EMPC) approach to selectively track regulation signals from an economic perspective. However, these methods rely on complex future prediction techniques. Moreover, the studies on day-ahead scheduling and real-time control are relatively isolated. The strategies for tracking regulation signals according to the reserve capacities remain to be further investigated.

Currently, only a few studies have focused on the frequency regulation potential of EV swapping mode. Among them [30], established a BSS scheduling model to participate in the regulation market at different time scales, with a particular emphasis on the busyness of the battery swapping service. A two-stage scheduling approach is employed in [31, 32] to formulate bidding strategies for BSS in the day-ahead regulation market. The intra-day uncertainties of swapping and regulation demand are carefully managed. The economic feasibility of BSS providing regulation services is verified in [33] by comparing the daily revenue and the long-term return on investment. However, the literature above focuses on ensuring that the bidding reserve capacities can cover BSS's worst operating scenario. The BSS's regulation strategy in real-time is neglected.

In summary, the current articles mainly focus on the frequency regulation of plug-in EVs. There is still a lack of research on the participation of BSS in the regulation market, especially the correlation between day-ahead scheduling and real-time control. Additionally, the uncertainties in the frequency regulation process are insufficiently considered. Stochastic programming [34,35] and robust optimization [36,37] are the most commonly used approaches to handle the uncertainty of regulation signals or EV states. However, stochastic

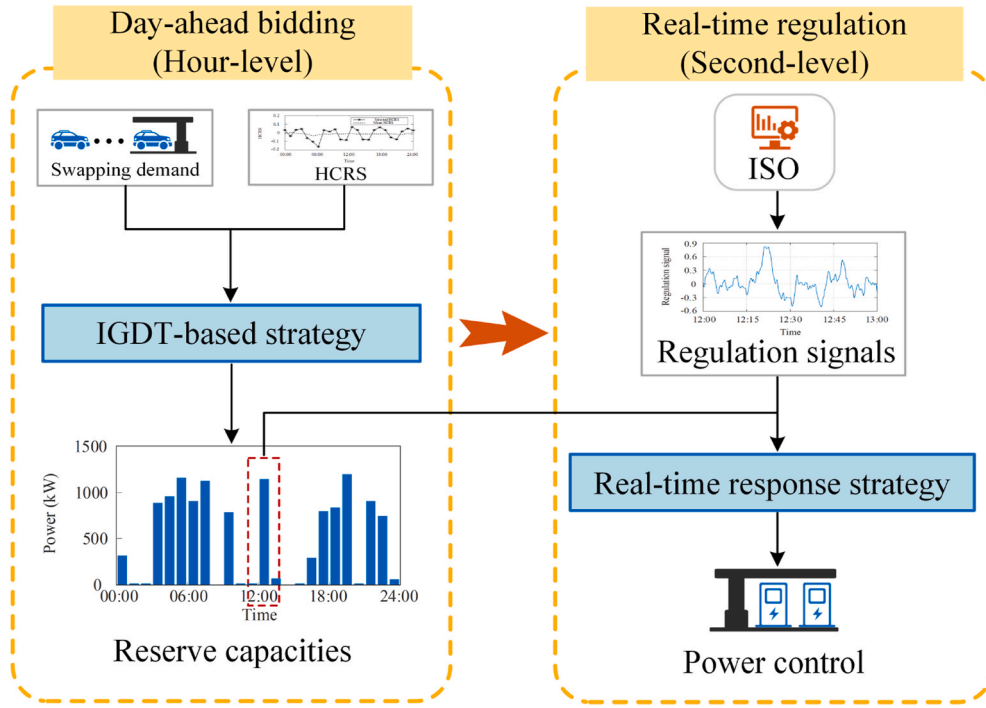


Fig. 1. Framework diagram of the proposed strategies for BSS participating in frequency regulation.

programming relies too much on prior probability distributions, while robust optimization often leads to overly conservative regulation reserves. Neural networks are used in [38,39] to establish mappings between state changes and frequency regulation strategies. However, their generalization ability needs improvement [40].

1.3. Main contributions

To address the above limitations, this study investigates strategies of BSS providing reserve capacities and real-time frequency regulation under the full consideration of uncertainties. The contributions of this study are as follows.

- (1) The uncertainties of swapping demand and regulation signals are carefully managed. In particular, a new method for establishing the uncertainty set of regulation signals is designed with the statistical analysis of PJM's historical data.
- (2) A day-ahead bidding strategy is proposed to obtain optimal reserve capacities leveraging the information-gap decision theory (IGDT), which achieves better performance than stochastic programming and robust optimization. A boundary autonomous selection (BAS) method is developed to solve the IGDT model, avoiding the assumption drawbacks of the traditional solving algorithm.
- (3) A real-time response strategy is proposed for BSS to track the regulation signals using the reserve capacities cleared in day-ahead market. The strategy can improve the overall economy of BSS while guaranteeing the frequency regulation performance.

2. Framework overview and model formulation

2.1. Framework overview

BSS participates in the frequency regulation of power system in two stages. Fig. 1 shows the framework diagram in this study. Firstly, BSS bids for reserve capacities in the day-ahead market. According to information like swapping demand, the proposed IGDT-based strategy can

generate the optimal reserve capacities for each hour of the following day. The next day, BSS provides real-time frequency regulation services during operation. Every few seconds, the regulation signal sent by ISO and the reserve capacity cleared by the day-ahead market are input into the proposed real-time response strategy. Considering the BSS's benefit, the strategy tracks the regulation signals online and produces the best response ratio. Then, BSS controls the charge/discharge power according to the response ratio to achieve frequency regulation.

With V2G technology, EV charging stations or BSS can behave like DGs (known as prosumers), which may cause voltage instability or congestion in the grid during operation. Many studies are based on the interaction between a single charging station or BSS and the power grid [41,42]. Some literature further adds network constraints to ensure that the energy or ancillary service schemes of multiple prosumers connected to different nodes are grid-friendly [43,44]. This paper mainly considers that a single BSS participates in ancillary service to provide support for solving the frequency instability of the power system. The focus is on dealing with the uncertainties that BSS faces during frequency regulation, ignoring the possible grid impact [31]. In further works, distribution network constraints can be added to verify the feasibility of bidding reserve capacities through power flow calculation.

2.2. BSS scheduling model

The generation and load are required to keep a dynamic balance during the operation of the power system. The grid frequency drops when the load exceeds generation; conversely, it rises. The ISO monitors the system frequency in real-time and sends regulation signals (a dimensionless number between -1 and 1) to the regulation resources. According to the signals, the resources adjust their power generation or consumption within the reserve capacities cleared in the day-ahead market. Thus, the system frequency is maintained around the rated value. Taking PJM as an example, the ISO sends the signal every 2 s. The signals filtered through a low-pass filter (i.e., RegA signals) by Area Control Error (ACE) are delivered to traditional regulation units. On the contrary, those filtered through a high-pass filter (i.e., RegD signals) are sent to dynamic regulation resources [45], such as BSS and photovoltaic.

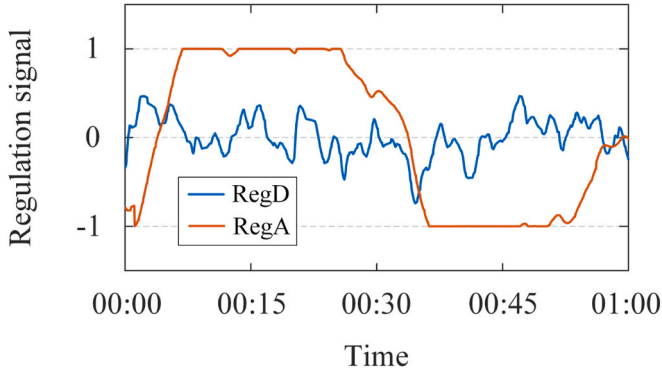


Fig. 2. Regulation signals within an hour from PJM.

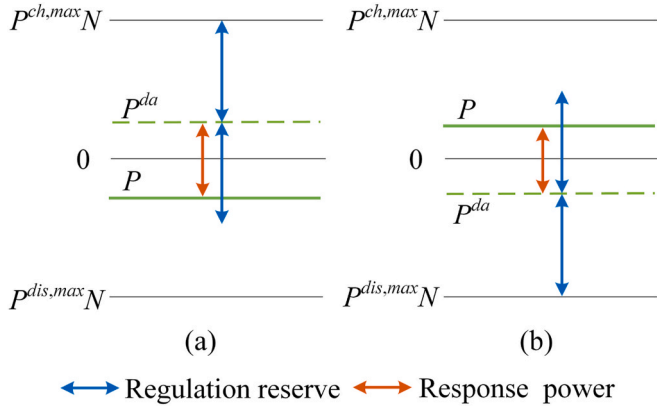


Fig. 3. Influence of regulation on the charging/discharging process. (a): Regulation up. (b): Regulation down.

The regulation signals within a certain hour in June 2023 are shown in Fig. 2. We can observe that RegD signals fluctuate much more violently than RegA signals and frequently cross 0. This is because the dynamic resources have a millisecond-level ramp rate, making them suitable for correcting minor frequency fluctuations.

Regulation resources like BSSs must participate in day-ahead bidding of the regulation market to determine the reserve capacities for each hour of the next day. To quantify the impact of frequency regulation on the hourly scale, the hourly cumulative regulation signals (HCRS) are calculated using all RegD signals within the same hour:

$$\pi_t = \sum_{\tau} \kappa_{\tau} \Delta \tau \quad \forall \tau \in \Omega_t \quad (1)$$

HCRS is also a dimensionless number between -1 and 1 . Fig. 3 illustrates the frequency regulation process during operation. This paper considers a common rule where the up and down reserve capacities are equal. As indicated by the blue arrows in Fig. 3, the reserve capacity cannot exceed BSS's maximum charging/discharging power. The maximum power is determined by both the chargers and inventory batteries within BSS. The constraints of reserve capacity are established below:

$$0 \leq R_t \leq P_t^{da} + P^{dis,max} N \quad (2)$$

$$0 \leq R_t \leq P^{ch,max} N - P_t^{da} \quad (3)$$

$$N = \min(N^{ch}, N^{ba}) \quad (4)$$

Secondly, HCRS is divided into positive and negative parts according to Equations (5) and (6), corresponding to regulation up and down. Regulation up indicates that ISO requires BSS to assist in raising the

system frequency. As a response, BSS reduces the charging power or increases the discharging power, as shown in Fig. 3(a). Regulation down requires BSS to act in reverse, as shown in Fig. 3(b).

$$\pi_t^+ = \max(0, \pi_t) \quad (5)$$

$$\pi_t^- = \min(0, \pi_t) \quad (6)$$

HCRS represents the proportion of regulation requirements in reserve capacities. However, BSS may not fully respond to HCRS due to scheduling and cost considerations. The actual response is constrained by:

$$0 \leq D_t^+ \leq \pi_t^+ R_t \quad (7)$$

$$0 \leq D_t^- \leq -\pi_t^- R_t \quad (8)$$

The BSS operation constraints are established below. The total power of BSS is the sum of each battery's charging/discharging power.

$$P_t = \sum_i P_{i,t} \quad (9)$$

According to Equations (5)–(8), BSS can either regulate up or down in the same hour. The baseline power and frequency regulation determines the actual power.

$$P_t = P_t^{da} - D_t \quad (10)$$

$$D_t = D_t^+ - D_t^- \quad (11)$$

Next, we formulate the energy constraints of the inventory batteries within the BSS. The energy inside battery i evolves according to Equation (12). $x_{i,t}$ is a binary variable indicating whether battery i is swapped at hour t . If so, $x_{i,t}$ is taken as 1. As a result, the energy difference between the fully-charged battery and the newly arrived one must be deducted from the energy indexed by i . Equation (15) limits the energy range. Constraints (12) and (15) jointly ensure that the batteries swapped to the EVs are fully charged. In addition, Equation (16) requires the BSS energy to be equal at the beginning and end of the day for operation stability.

$$E_{i,t} = E_{i,t-1} + P_{i,t-1}^{ch} \eta^{ch} - P_{i,t-1}^{dis} / \eta^{dis} - x_{i,t} (\bar{E} - E) \quad (12)$$

$$P_{i,t}^{ch} = \max(0, P_{i,t}) \quad (13)$$

$$P_{i,t}^{dis} = \max(0, -P_{i,t}) \quad (14)$$

$$\underline{E} \leq E_{i,t} \leq \bar{E} \quad (15)$$

$$\sum_i E_{i,0} = \sum_i E_{i,T} \quad (16)$$

The number of served EVs depends on the swapping demand and swapping facilities. EVs that fail to get services immediately will queue up, affecting service quality. In the next time slot, they will accumulate to the newly generated swapping demand:

$$\sum_i x_{i,t} \leq \min(\tilde{N}_t^{sw} + QUE_{t-1}, N^{sf}) \quad (17)$$

$$QUE_t = \tilde{N}_t^{sw} + QUE_{t-1} - \sum_i x_{i,t} \quad (18)$$

BSS provides frequency regulation services to seek more profits. The objective function maximizes the revenue from the regulation market and swapping service. Meanwhile, the energy cost and the punishment for frequency regulation shortage are minimized. The battery degradation is more significant in long-term planning as it causes a negligible effect in one day and increases the nonlinearity of the model [46]. Therefore, the impact of degradation is not considered in the model.

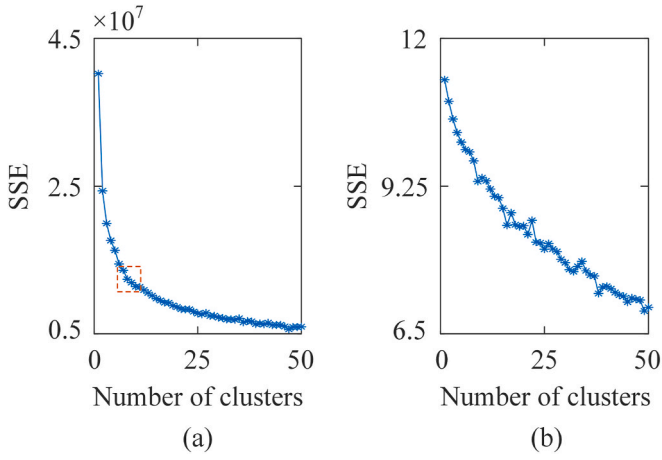


Fig. 4. The relationship between SSE and the cluster numbers. (a): Wind power generation. (b): HCRS.

$$\max F = F^{\text{reg}} + F^{\text{sw}} - F^{\text{en}} - F^{\text{pun}} \quad (19)$$

$$F^{\text{reg}} = \sum_t \lambda_t^{\text{rc}} R_t + \lambda_t^{\text{res}} (D_t^+ + D_t^-) \quad (20)$$

$$F^{\text{sw}} = \sum_t \sum_i \lambda_i^{\text{sw}} x_{i,t} - \sum_t \lambda^{\text{que}} \text{QUE}_t \quad (21)$$

$$F^{\text{en}} = \sum_t \lambda_t^{\text{grid}} P_t^{\text{da}} \quad (22)$$

$$F^{\text{pun}} = \lambda^{\text{pun}} \sum_t \pi_t^+ R_t - D_t^+ - \pi_t^- R_t - D_t^- \quad (23)$$

3. Uncertainty analysis

The uncertainties considered in this paper include battery swapping demand and HCRS. Similar to the arrival of plug-in EVs, the swapping demand is generally assumed to follow the Poisson distribution [47] to generate typical scenarios. However, the regulation signals are complex and difficult to process. Therefore, we mainly analyze the HCRS uncertainty in this section.

3.1. Statistical analysis of regulation signal

HCRS is usually handled by scenario-based method [48], prediction [29], or even treated as fixed values [49]. Actually, HCRS is hard to predict or represent by typical scenarios because of the high randomness of dynamic balance between generation and load.

For verification, we cluster the historical data of wind power generation and HCRS using the K-means method. Both datasets include 365 samples in a year. We sequentially cluster them into 1 to 50 clustering centers and calculate the sum of squared errors (SSE). Then, the elbow method is used to assess the clustering effectiveness. The optimal number of clusters is the point with the largest curvature in the SSE curve, similar to the elbow in a curved arm. Fig. 4 compares the SSE curves of wind power and HCRS. As shown in Fig. 4(a), the curvature of the wind power in the red box is significantly larger than that at other positions. In Fig. 4(b), the HCRS curve maintains a relatively consistent curvature regardless of the cluster numbers. It indicates that wind power can be effectively represented by several typical scenarios, whereas HCRS cannot.

According to Equation (1), HCRS can be considered as the integral of all regulation signals in 1 h. It reflects the impact of the frequency regulation on energy over an hour. To quantify the impact over a longer time, the cumulative regulation signals (CRS) are defined as:

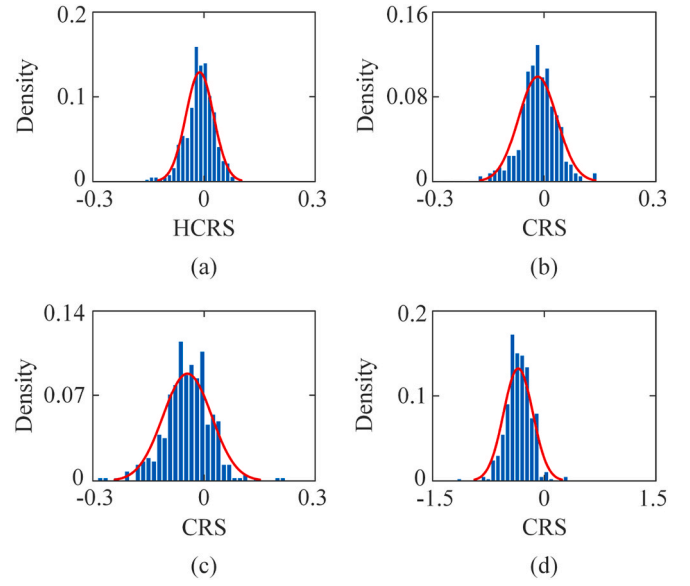


Fig. 5. Histograms and normal distribution fitting of HCRS and CRS. (a): HCRS for the first hour of the day. (b): CRS for the first 3 h (c): CRS for the first 5 h (d): CRS for 24 h.

$$\pi_t = \sum_{\tau} \kappa_{\tau} \Delta \tau \quad \forall \tau \in [\Omega_t, \Omega_{t+1}, \dots, \Omega_{t+T-1}] \quad (24)$$

Statistical analysis is conducted based on PJM's historical data within a year. The results shown in Fig. 5 include the HCRS for the first hour, as well as the CRS for the first three, five, and 24 h of the day. Fig. 5 (a) suggests that more than 99.52% of HCRS is distributed within the range of $[-0.15, 0.15]$. The samples gather around zero and exhibit a light-tailed characteristic. By comparing Fig. 5(a)–5(c), we can identify that the impact of regulation signals on energy accumulates slowly over time; moreover, the mean value gradually shifts to the left, indicating that there are slightly more negative signals than positive ones. As depicted in Figs. 5(d), 97.53% of CRS for 24 h fall within the range of $[-0.7, 0.3]$. It indicates that although regulation signals fluctuate randomly, the positive and negative impacts offset each other. This characteristic is also reflected in HCRS at the hour-level.

Through the above analysis, the features of HCRS under the RegD signals can be summarized as follows.

- (1) HCRS cannot be represented by several typical scenarios;
- (2) Different from common uncertainties like wind power, HCRS can be positive or negative;
- (3) The numbers of positive and negative HCRS in a day are close;
- (4) HCRS values are relatively concentrated, while their expectation is exceptionally close to zero.

3.2. Uncertainty modeling

According to the feature (1) mentioned above, the scenario-based methods are abandoned in this paper. First, the envelope bound model is employed to describe the uncertainty of swapping demand, as captured by:

$$\begin{cases} \frac{|N_t^{\text{sw}} - \tilde{N}_t^{\text{sw}}|}{\tilde{N}_t^{\text{sw}}} \leq \alpha^{\text{sw}} \\ N_t^{\text{sw}} \in \mathcal{N} \end{cases} \quad (25)$$

where $\mathcal{N}$ is the set of natural numbers and α is the uncertainty radius. Due to features (2) and (4), the envelope bound model does not apply to HCRS. Therefore, we design an uncertainty set within a limited range for

HCRS. Firstly, we calculate the maximum up/down fluctuation of HCRS in hour t on the k th day

$$\bar{\pi}_{k,t} = \max_{\omega} \sum_{\tau=1}^{\omega} \kappa_{\tau} \Delta \tau \quad \forall \tau \in \Omega_t \quad (26)$$

$$\underline{\pi}_{k,t} = \min_{\omega} \sum_{\tau=1}^{\omega} \kappa_{\tau} \Delta \tau \quad \forall \tau \in \Omega_t \quad (27)$$

We compare the maximum fluctuation of different days. The maximum and minimum values are taken as the upper and lower bounds of HCRS, as defined by Equations (28) and (29). They represent the greatest impact that frequency regulation can have on energy in hour t .

$$\bar{\pi}_t = \max_k \bar{\pi}_{k,t} \quad (28)$$

$$\underline{\pi}_t = \min_k \underline{\pi}_{k,t} \quad (29)$$

Next, the expectation of HCRS over K days is calculated by:

$$\tilde{\pi}_t = \frac{\sum_k \bar{\pi}_{k,t}}{K} \quad (30)$$

We make HCRS increase or decrease proportionally within $[\bar{\pi}_t, \underline{\pi}_t]$, with $\tilde{\pi}_t$ as the center. This resolves the issues brought about by features (2) and (4). The uncertainty sets of swapping demand and HCRS are finally constructed as

$$\begin{cases} U(\alpha^{sw}, \tilde{N}_t^{sw}) = \left\{ N_t^{sw} \in \mathcal{N} : \frac{|N_t^{sw} - \tilde{N}_t^{sw}|}{\tilde{N}_t^{sw}} \leq \alpha^{sw} \right\} \\ U(\alpha^{\pi}, \tilde{\pi}_t) = \{ \pi_t : \tilde{\pi}_t - \alpha^{\pi}(\tilde{\pi}_t - \underline{\pi}_t) \leq \pi_t \leq \tilde{\pi}_t + \alpha^{\pi}(\bar{\pi}_t - \tilde{\pi}_t) \} \\ \alpha^{sw} \geq 0, 0 \leq \alpha^{\pi} \leq 1 \end{cases} \quad (31)$$

4. IGDT-based bidding strategy

In this section, we utilize the established BSS model and uncertainty sets to develop an IGDT-based strategy. The IGDT method is an interval optimization technique. Since this method makes no assumptions about the probability distribution of uncertain variables, it is particularly suitable for situations where a consistent probability distribution is unavailable. First, we establish the IGDT model and convexify bilinear terms. Then, we propose the BAS method to solve the strategy considering the characteristics of uncertainties.

4.1. IGDT modeling and linearization

Based on Equation (31), the uncertainty radii α^{sw} and α^{π} are unified into a weighted sum form:

$$\alpha = \lambda^{sw} \alpha^{sw} + \lambda^{\pi} \alpha^{\pi} \quad (32)$$

α can be used to measure the widths of gaps between prediction (i.e., $\tilde{N}_t^{sw}$ and $\tilde{\pi}_t$) and uncertainties. When $\alpha = 0$, the uncertainty set compresses to the prediction. At this point, the model becomes a deterministic optimization based on the prediction. As α grows, the uncertainty range gradually expands, leading to an increase in the feasible region and variability of the objective function.

BSS participates in the regulation market primarily to acquire more profit. Therefore, the opportunity function is considered here to determine how the BSS can benefit from the potential variations in uncertainties. The IGDT-based scheduling model is established as the following bi-level form:

$$\begin{cases} \min \alpha \\ s.t. \begin{cases} \max F \geq (1 + \sigma) F^b \\ N_t^{sw} \in U(\alpha^{sw}, \tilde{N}_t^{sw}), \pi_t \in U(\alpha^{\pi}, \tilde{\pi}_t) \\ (1) - (23), (26) - (32) \\ F^b = F|_{N_t^{sw} = \tilde{N}_t^{sw}, \pi_t = \tilde{\pi}_t} \end{cases} \end{cases} \quad (33)$$

where F^b is the reference value obtained by deterministic optimization when $\alpha = 0$; σ denotes the opportunity level factor, which is a parameter set by operators according to the willingness of risk speculation. The model aims to find the minimum α that can increase the profit to a satisfaction level.

Note that $\pi_t^+ R_t$ and $\pi_t^- R_t$ in Equations (7) and (8) turn into bilinear terms since HCRS become variables in the IGDT model. They make the model non-convex and bring difficulties to the solution. We linearize them to construct convex functions. It is observed that each bilinear term is the product of two continuous variables. In particular, the positive HCRS (or the inverse of negative HCRS) is between 0 and 1. The quasi-precision method can be precisely used to handle such a form. Therefore, we introduce this method to transform the bilinear constraints into mixed integer linear constraints.

The core idea of the quasi-precision method is to represent the variables between 0 and 1 as a series of binary variables, that is, $x = \sum_{q=0}^Q 2^{-q} d_q + \delta$, where Q is the number of binary variables and δ is the error belonging to $[0, 2^{-Q}]$. Apparently, the error drops as Q increases. Taking the positive HCRS as an example, it can be expressed as:

$$\pi_t^+ = \sum_{q=0}^Q 2^{-q} d_{q,t}^+ + \delta_t^+ \quad (34)$$

Accordingly, Equation (7) can be reformulated as:

$$\begin{cases} D_t^+ \geq 0 \\ D_t^+ \leq \left(\sum_{q=0}^Q 2^{-q} d_{q,t}^+ + \delta_t^+ \right) R_{i,t} \end{cases} \quad (35)$$

Note that the right side of the second inequality still contains two nonlinear terms. The first term $d_{q,t}^+ R_t$ is the product of a binary variable and a continuous variable. It is equivalently transformed by the McCormick Envelope method:

$$\begin{cases} \zeta_{q,t}^+ \leq d_{q,t}^+ \bar{R}_t \\ \zeta_{q,t}^+ \geq d_{q,t}^+ \underline{R}_t \\ \zeta_{q,t}^+ \leq \bar{R}_t - \underline{R}_t (1 - d_{q,t}^+) \\ \zeta_{q,t}^+ \geq \bar{R}_t - \bar{R}_t (1 - d_{q,t}^+) \end{cases} \quad (36)$$

where $\zeta_{q,t}^+$ is the auxiliary variable. $\bar{R}_t = (p^{ch,max} + p^{dis,max})N$ and $\underline{R}_t = 0$ are the upper and lower bounds of reserve capacity.

To address the second bilinear term, an auxiliary variable $\beta_{q,t}^+$ is created to represent $\delta_t^+ R_t$. Since we have $\delta_t^+ \leq 2^{-Q}$, the term can be replaced by adding constraint $0 \leq \beta_{q,t}^+ \leq 2^{-Q} R_t$.

Finally, Equation (7) is rewritten as the following series of linear constraints:

$$\left\{ \begin{array}{l} 0 \leq D_t^+ \leq \sum_{q=0}^Q 2^{-q} \zeta_{q,t}^+ + \beta_{q,t}^+ \\ \pi_t^+ = \sum_{q=0}^Q 2^{-q} d_{q,t}^+ + \delta_t^+ \\ 0 \leq \delta_t^+ < 2^{-Q} \\ \zeta_{q,t}^+ \leq d_{q,t}^+ \bar{R}_t \\ \zeta_{q,t}^+ \geq d_{q,t}^+ \underline{R}_t \\ \zeta_{q,t}^+ \leq R_t - \underline{R}_t (1 - d_{q,t}^+) \\ \zeta_{q,t}^+ \geq R_t - \bar{R}_t (1 - d_{q,t}^+) \\ 0 \leq \beta_{q,t}^+ \leq 2^{-Q} R_t \end{array} \right. \quad (37)$$

$\pi_t^- R_t$ in Equation (8) is linearized by the same method.

When determining the reserve capacity, the possible impact of both regulation up and down to BSS energy must be considered. BSS must reserve surplus energy to accommodate frequency regulation in any direction. The upper and lower bounds of HCRS obtained in Section 3.2 are used to add the following energy constraints to the model:

$$\underline{EN}^{ba} \leq E_t^{bss} + \max(P_t^{da} - \bar{\pi}_t R_t, 0) \eta^{ch} - \max(\bar{\pi}_t R_t - P_t^{da}, 0) / \eta^{dis} \leq \bar{EN}^{ba} \quad (38)$$

$$\underline{EN}^{ba} \leq E_t^{bss} + \max(P_t^{da} - \underline{\pi}_t R_t, 0) \eta^{ch} - \max(\underline{\pi}_t R_t - P_t^{da}, 0) / \eta^{dis} \leq \bar{EN}^{ba} \quad (39)$$

4.2. Solving method

The scheduling model (33) belongs to a two-layer optimization problem, which is difficult to solve directly. Due to the nesting nature (i.e., $U(\alpha^1, \tilde{u}) \subseteq U(\alpha^2, \tilde{u})$ if $\alpha^1 \subseteq \alpha^2$), the solution will find the lower and upper bounds of the uncertainty radius [50]. As the common practice to solve the IGDT model, all these boundaries must be assumed to be higher or lower than the predicted value simultaneously. However, this situation rarely happens in reality. As feature (3) indicates, the number of HCRS on both sides of the predicted value is close. Furthermore, unlike the DG generation, it is impossible to intuitively judge whether taking the upper or lower bound benefits the objective function for each hour. To address the above two issues, we propose a BAS method to solve the IGDT model. The method first takes a set of binary variables to select the upper and lower bounds by optimization. The aim is to transform the IGDT model (33) into a single-layer optimization problem. Afterward, the quantitative range of the selected upper and lower bounds is constrained with the statistics of HCRS to conform to feature (3). Finally, the IGDT model is reformulated as a mixed integer linear programming (MILP) form, which can be solved directly by commercial solvers.

In the BAS method, the upper or lower bound is selected when $\bar{y}_t$ or $\underline{y}_t$ takes 1. As for the swapping demand, we have

$$N_t^{sw} = \underbrace{\bar{y}_t (1 + \alpha^{sw}) \tilde{N}_t^{sw}}_{\text{Upper bound}} + \underbrace{\underline{y}_t (1 - \alpha^{sw}) \tilde{N}_t^{sw}}_{\text{Lower bound}} \quad (40)$$

$$\bar{y}_t + \underline{y}_t = 1 \quad (41)$$

Equation (40) can be simplified to the following form after substituting Equation (41). The meaning is to make the boundary rise or fall from the predicted value with the amplitude of uncertainty radius.

$$N_t^{sw} = \underbrace{\tilde{N}_t^{sw}}_{\text{Predicted value}} + \underbrace{\bar{y}_t \alpha^{sw} \tilde{N}_t^{sw}}_{\text{Rise}} - \underbrace{\underline{y}_t \alpha^{sw} \tilde{N}_t^{sw}}_{\text{Fall}} \quad (42)$$

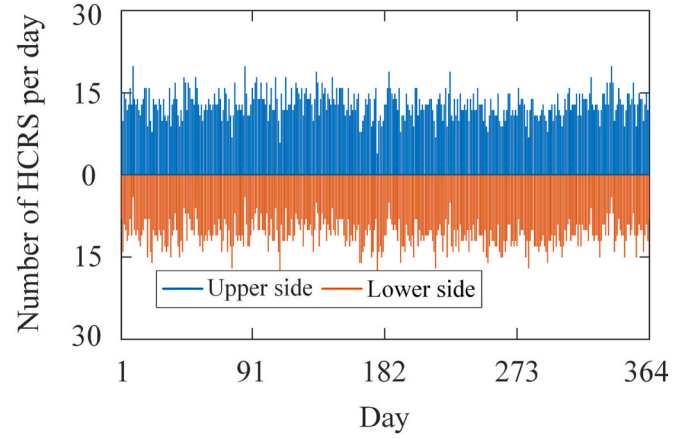


Fig. 6. Number of HCRS on both sides of the mean value per day within a year.

Similarly, a set of binary variables is added to HCRS to choose the bounds, which is

$$\pi_t = \tilde{\pi}_t + \bar{z}_t \alpha^\pi (\pi_t - \tilde{\pi}_t) - \underline{z}_t \alpha^\pi (\pi_t - \underline{\pi}_t) \quad (43)$$

$$\bar{z}_t + \underline{z}_t = 1 \quad (44)$$

Based on the above two equations, the expanded form of Equation (43) is:

$$\pi_t = \tilde{\pi}_t + \bar{z}_t \alpha^\pi \bar{\pi} - \alpha^\pi \tilde{\pi}_t + \underline{z}_t \alpha^\pi \underline{\pi}_t \quad (45)$$

Fig. 6 presents the number of HCRS on both sides of the mean value per day within a year. It can be observed that there are no instances where all HCRS values are exclusively above or below the mean value. Meanwhile, there are slightly more HCRS on the upper side. We limit the number of HCRS on either side (i.e., $\sum_t \bar{z}_t$ or $\sum_t \underline{z}_t$) to a specific range to make the model closer to real-world conditions. Specifically, based on the statistics from Fig. 6, we sort the numbers of HCRS on the upper side per day (i.e., the blue bars). The upper and lower bounds of $\sum_t \bar{z}_t$ are set as the first quartile Q_1 and the third quartile Q_2 , which is

$$Q_1 \leq \sum_t \bar{z}_t \leq Q_2 \quad (46)$$

Since the total number of HCRS per day remains constant, establishing constraints for HCRS on the lower side is unnecessary. This approach addresses the issue posed by feature (3).

Equations (42) and (45) also contain bilinear terms consisting of binary and continuous variables. Similar to Equation (36), the McCormick Envelope method equivalently transforms them into linear constraints. For example, $\bar{z}_t \alpha^\pi$ is replaced by introducing an auxiliary variable $\bar{Z}_t$ and the following new constraints:

$$\left\{ \begin{array}{l} \bar{Z}_t \leq \bar{\alpha}^\pi \bar{z}_t \\ \bar{Z}_t \geq \underline{\alpha}^\pi \bar{z}_t \\ \bar{Z}_t \leq \alpha^\pi - \bar{\alpha}^\pi (1 - \bar{z}_t) \\ \bar{Z}_t \geq \alpha^\pi - \underline{\alpha}^\pi (1 - \bar{z}_t) \end{array} \right. \quad (47)$$

where $\bar{\alpha}^\pi$ and $\underline{\alpha}^\pi$ is the upper and lower limits of α^π , taken as 0 and 1, respectively. The same method is applied to $\underline{z}_t \alpha^\pi$, $\bar{y}_t \alpha^{sw}$ and $\underline{y}_t \alpha^{sw}$.

With the BAS method described above, the IGDT model is finally transformed into the following MILP form:

$$\left\{ \begin{array}{l} \min \alpha \\ \text{s.t.} \left\{ \begin{array}{l} F \geq (1 + \sigma)F^b \\ (1) - (6), (9) - (23), (26) - (32), (37) - (39), (41), (44), (46) - (47) \\ N_t^{sw} = \bar{Y}_t \tilde{N}_t^{sw} - \underline{Y}_t \tilde{N}_t^{sw} + \tilde{N}_t^{sw} \\ \pi_t = \tilde{\pi}_t + \bar{Z}_t \tilde{\pi}_t + \underline{Z}_t \tilde{\pi}_t - \alpha \tilde{\pi}_t \\ F^b = F|_{N_t^{sw} = \tilde{N}_t^{sw}, \pi_t = \tilde{\pi}_t} \end{array} \right. \end{array} \right. \quad (48)$$

5. Real-time response strategy

Based on the reserve capacities cleared in the day-ahead market, the regulation resources need to follow the regulation signals delivered by ISO every 2 s. Different ancillary markets have established various indexes to evaluate real-time regulation performance. BSS has a natural advantage in ramp rate as the chargers can control power within milliseconds. Therefore, response accuracy is the critical factor affecting regulation performance.

The intra-day HCRS is likely to differ significantly from the one in the day-ahead optimization. Many studies focused on optimizing reserve capacities while neglecting the real-time response. The real-time regulation signals exhibit high randomness during operation. The response to each signal changes power briefly for a few seconds. The full response to regulation signals does not necessarily represent the overall economic optimality of regulation resources [29]. As the regulation signals fluctuate rapidly, the resources must determine the response ratio to make a trade-off between regulation performance and operation benefits. The response ratio is defined as the proportion of the frequency regulation deployment to the requirement. For this reason, we propose a real-time response strategy suitable for dynamic resources.

The strategy is based on lightweight quadratic programming and is executed every h seconds (h can be set by the operators). The decision variable is the response ratio γ to the regulation signal. From the day-ahead scheduling results, the optimal power variation caused by frequency regulation at hour t (i.e., D_t) can be directly obtained. Ideally, if the actual power variation of the hour exactly matches D_t , it is the most advantageous for BSS's interests. Therefore, the real-time scheduling aims to minimize the square of the difference between the two aspects using the cleared reserve capacity R_t . In addition, the relaxation variable μ is penalized to ensure that the strategy is feasible in extreme

conditions. The details will be further introduced in constraints (52) and (53). For the optimization of the τ -th signal in hour t , the objective function is expressed as:

$$\min \left[D_t - \left(\sum_{\tau'=1}^{\tau-1} \gamma_{\tau'} \kappa_{\tau'} \Delta \tau + \sum_{\tau'=\tau}^T \gamma_{\tau'} f(\tau') \kappa_{\tau'} \Delta \tau \right) R_t \right]^2 + M(\mu_1 + \mu_2) \quad (49)$$

where T represents the total number of regulation signals within 1 h (e.g., T is 1800 for PJM), and M denotes a large positive number. The optimization model takes the time horizon as 1 h and takes all T regulation signals into consideration. The challenging issue is that the regulation signals from $\tau+1$ to T are unknown. Although the fluctuation of regulation signals is difficult to capture, the variation trend can be generalized through the current signal and statistics to reduce the influence of prediction error. The future regulation signals change from the current signal. In the long run, the mean value of the RegD signal is essentially zero, which can be illustrated from the statistics in Fig. 7. Based on these two points, the exponential decay signal f is introduced and multiplied by the current regulation signal to represent the predicted values of future signals. This term is widely used to characterize the attenuation process of signal strength [51]. Thus, the future signals can keep the current trend and gradually decay to zero. Statistical results for PJM signals indicate that RegD signals cross zero every 122 s on average. As shown in Equation (50), this paper sets the base as 0.95, allowing the signal to decay by over 95% within 122 s.

$$f(\tau') = 0.95^{\tau'-\tau} \quad (50)$$

The optimization must satisfy the following constraints. To prevent over-charging/over-discharging caused by continuous and prolonged frequency regulation in the same direction, BSS still needs to follow energy constraints, as shown in Equation (51). The range $[\tau, \tau+0.5h-1]$ represents the set of all regulation signals before the next optimization (PJM sends signals every 2 s so that each optimization covers $0.5h$ signals). Equation (52) sets a lower bound for the response ratio to each regulation signal. Equation (53) requires that the average response ratio for the latest j signals be kept at a high level to ensure a good tracking performance. The lower bound γ^{ave} can be adjusted by BSS operators based on expectations for regulation performance ($\gamma^{ave} = 1$ means full response). When the BSS energy is almost full or empty, it may be necessary to maintain a low response ratio to satisfy the energy constraint in Equation (51). To avoid conflict with the set lower bounds, relaxation variables μ_1 and μ_2 are added to Equations (52) and (53) and penalized in the objective function. The purpose is to make the response ratio strictly obey the lower bounds during normal circumstances and have a feasible solution in extreme conditions.

$$\text{s.t. } 0 \leq E_t^{bss} + \sum_{\tau' \in [\tau, \tau+0.5h-1]} \gamma_{\tau'} \kappa_{\tau'} \Delta \tau \leq \bar{E} N^{ba} \quad (51)$$

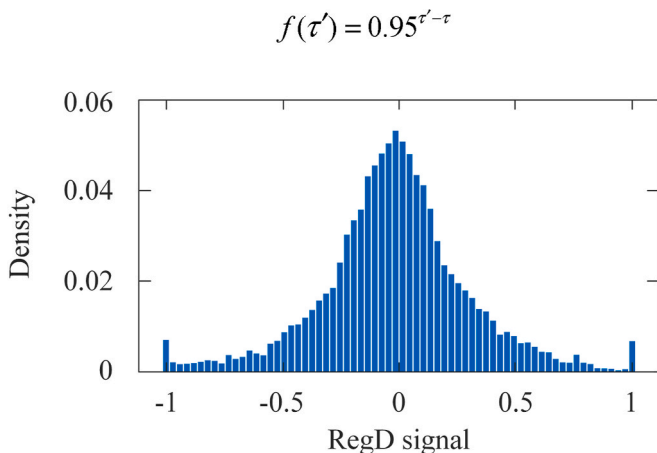


Fig. 7. Histogram of RegD signals.

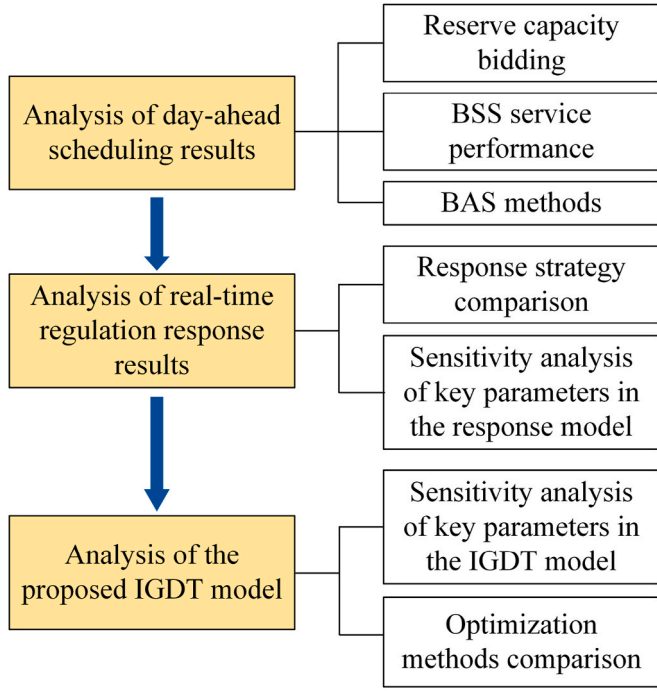


Fig. 8. Overall flowchart of the experimental case design.

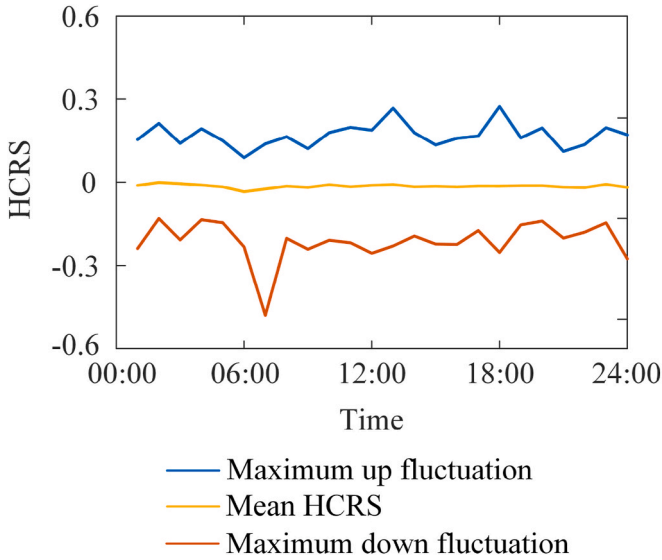


Fig. 9. Mean value and maximum up/down fluctuation of HCRS.

$$\begin{cases} \gamma - \mu_1 \leq \gamma_\tau \leq 1 \\ \mu_1 \geq 0 \end{cases} \quad (52)$$

$$\begin{cases} \gamma_\tau^{ave} \geq \gamma^{ave} - \mu_2 \\ \gamma_\tau^{ave} = \frac{\sum_{t=\tau-j+1}^{\tau} \gamma_t}{j} \\ \mu_2 \geq 0 \end{cases} \quad (53)$$

The strategy can make actual power approximate the day-ahead optimization results while considering the response ratio. Due to the low dimensions of decision variables, the solution speed is extremely fast.

The pseudocode for the day-ahead bidding strategy and the real-time

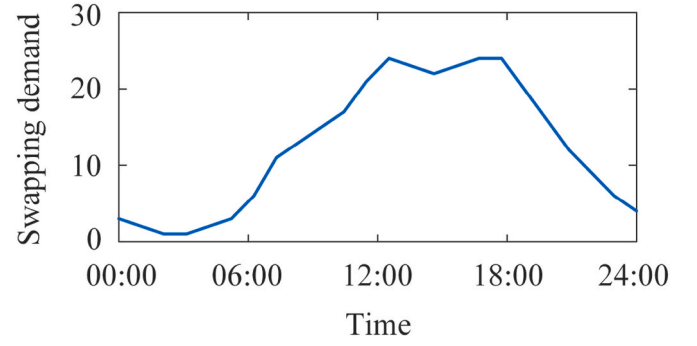


Fig. 10. Forecast profile of swapping demand.

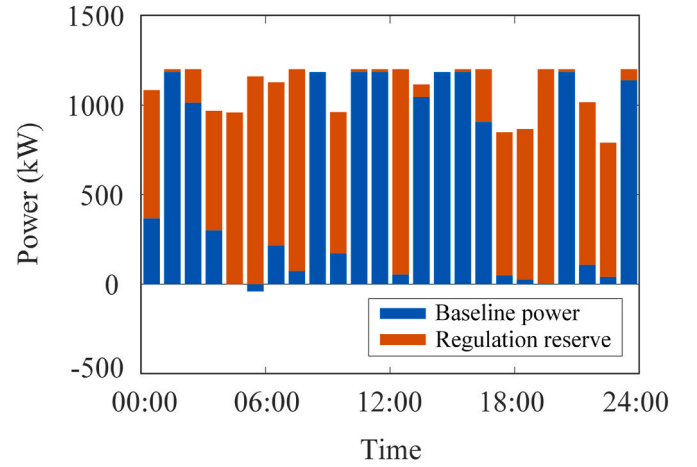


Fig. 11. Optimal regulation reserve capacity and baseline power.

response strategy is shown in Algorithm 1.

Algorithm 1. The pseudocode for the day-ahead bidding strategy and the real-time response strategy

-
- 1: Input parameters. Input the predicted value of swapping demand and HCRS.
 - 2: Solve the deterministic optimization problem (19) to obtain F^b .
 - 3: Input F^b . Solve the problem (48) to obtain optimal R_t and D_t for each hour.
 - 4: for $t = 1, 2, \dots, 24$ do
 - 5: Input R_t and D_t .
 - 6: for $\tau = 1, 2, \dots, T$ do
 - 7: Input the first to τ -th regulation signals at hour t . Solve the problem (48) to obtain optimal γ .
 - 8: end for
 - 9: end for
-

6. Case studies

In this section, we verify the proposed strategy by simulation. Fig. 8 illustrates the overall flowchart of the experimental case design. First, we analyze the day-ahead scheduling results to verify the performance of the bidding plan, battery swapping service, and BAS solving method. Then, the real-time regulation response results are validated by comparing different response strategies and analyzing key parameters. Finally, the uncertainty processing capability of the proposed IGDT model is tested by conducting sensitivity analysis and method comparisons.

6.1. Parameter settings

The simulation is implemented using the Cplex solver and Yalmip toolbox in MATLAB R2022a. The hardware is a laptop with a Core i7-

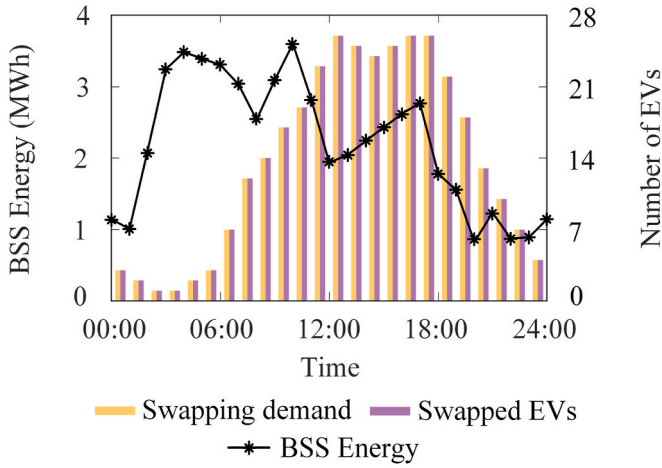
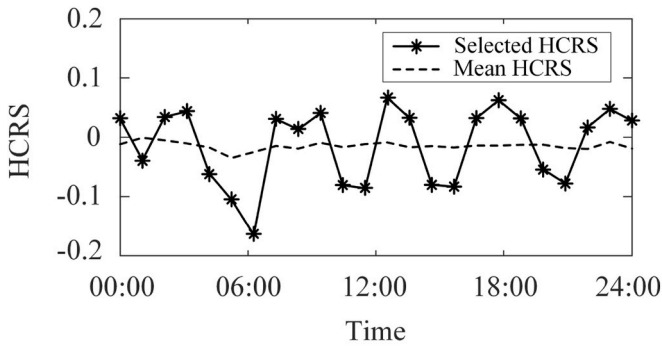
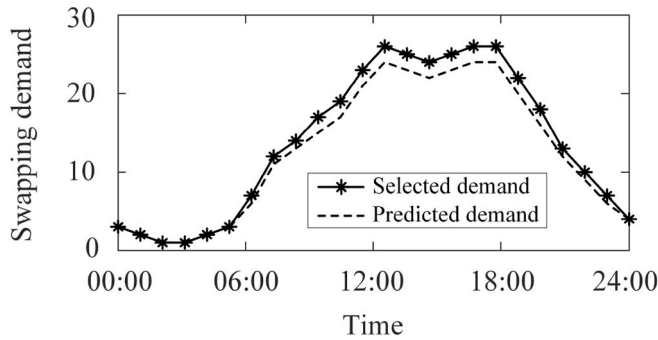


Fig. 12. Variation of BSS energy and swapping EVs.



(a)



(b)

Fig. 13. Profiles of the boundaries selected by the BAS method. (a): HCRS. (b): Swapping demand.

12700H CPU, 16 GB RAM, and a 500 GB hard drive. The regulation signals from PJM open source data [52] range from July 2022 to June 2023. Fig. 9 shows the generated HCRS parameters. Electricity and reserve capacity prices refer to the literature [31], which is taken from the historical data of PJM. In this paper, the BSS parameters are set considering the actual situation, developing trends and other literature. Due to the lack of public data on swapping demand, the visiting data from gas stations are adopted instead in some studies. In our simulation, the hourly visiting probability in [53] is used to generate the predicted swapping demand shown in Fig. 10. BSS has 30 chargers with a rated

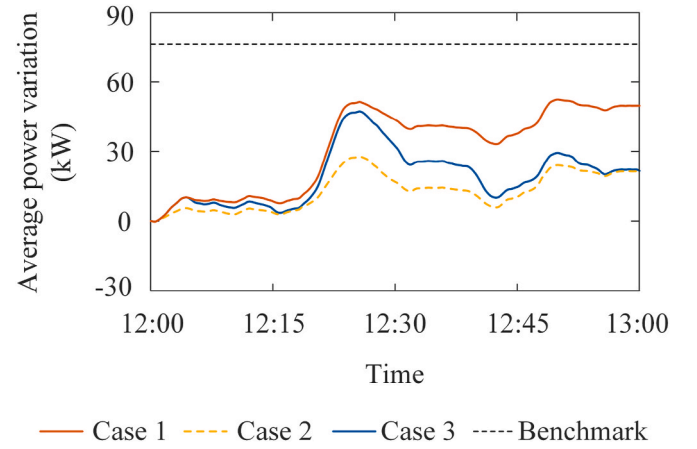


Fig. 14. Evolution of average power variation between 12:00 and 13:00.

charging/discharging power of 40 kW and an efficiency of 0.95. It conforms to the actual engineering as the BSS generally uses low charging/discharging power to extend the battery life. BSS is set to reserve 80 inventory batteries, which is in the same order of magnitude as the actual scale. There are 20 full-charged batteries and 60 depleted batteries at the beginning of the day, with the SOC of 0.95 and 0.1, respectively. The battery capacity is set as 60 kWh, consistent with the swapping EVs such as NIO ONVO L60. The Q in the quasi-precision method is taken as 10. The opportunity level factor is set to 0.2. The weights of uncertainty radii are both 0.5. Real-time optimization is performed every 30 s with a minimum response ratio of 0.5. The lower bound of the average response ratio per 10 min is set to 0.7.

6.2. Day-ahead scheduling results

To quantify the results obtained by the proposed day-ahead bidding strategy, the reserve capacity and baseline power are presented in Fig. 11. The baseline power takes into account the electricity prices and the state of inventory batteries. On average, 80.44% of the capacity left by the baseline power is used for regulation reserve. BSS provides more reserves from 03:00 to 08:00 and 17:00 to 20:00. In the first period, most inventory batteries are full, leading to low charging power. In the second period, the electricity price is high. BSS reduces the baseline power to obtain higher profits in the regulation market.

Next, we analyze the BSS operational states. Fig. 12 shows the variation of BSS energy and swapping service conditions. Evidently, the BSS energy remains constant at the beginning and end of the day. BSS energy has not reached the upper or lower limits throughout the day, leaving space for possible up or down regulation. Additionally, all EVs immediately swap their batteries upon arrival, and no queuing occurs. It indicates that the proposed method effectively considers BSS's service quality.

Fig. 13 shows the HCRS values and swapping demands selected by the IGDT method. As depicted in Fig. 13(a), there are 14 positive HCRS and 10 negative ones. Most of the values are within $[-0.1, 0.1]$, which accords with the statistical results in Fig. 5(a). It validates the practicability of the scenario considered by the proposed BAS method. As shown in Fig. 13(b), all the selected swapping demands are on the upper side of the predicted value. This is because more swapping EVs can bring higher service income within the support capability of the fully-charged batteries.

6.3. Real-time regulation response

Next, we analyze the performance of the proposed real-time response strategy. As discussed in Section 5, D_t in day-ahead scheduling results represents the optimal power variation caused by frequency regulation

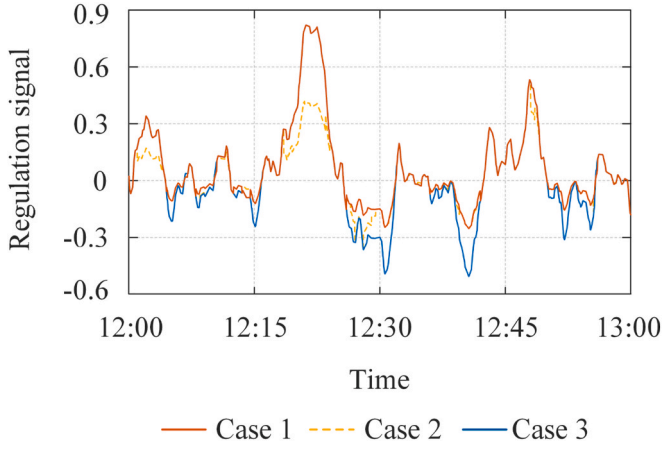


Fig. 15. Response ratios to regulation signals between 12:00 and 13:00.

Table 1

Comparison of the intra-day performances under different lower bounds of the average response ratio.

γ_{τ}^{ave}	0.6	0.7	0.8	0.9	1
Intra-day revenue (\$)	482.51	469.73	444.39	425.9	403.33
Average difference with benchmark (%)	31.94	34.81	43.12	50.18	62.99
Actual average response ratio	0.75	0.8	0.85	0.92	1

under ideal conditions. We use D_t as the benchmark and compare it to the actual power variation. The following three cases are set up for performance comparison.

Case 1. Regulation signals are responded to using the proposed real-time strategy.

Case 2. Regulation signals are responded to using the proposed real-time strategy; however, the exponential decay signal is removed from the objective function (49).

Case 3. All frequency regulation signals are fully responded to.

Note that the response to the regulation signals causes the power to change every few seconds; however, as day-ahead optimization is formulated on the hourly scale, D_t reflects the average power variation over hour t (i.e. $D_t = \sum_{\tau=1}^T \gamma_{\tau} \kappa_{\tau} R_t \Delta \tau$). Fig. 14 illustrates the evolution of average power variation between 12:00 and 13:00. The variation starts from zero at 12:00 since there is no response to any signal yet. As for Case 3, the variation gradually diverges from the benchmark of 76.35 kWh after 12:25 and turns out to be 21.57 kWh at 13:00. It illustrates a significant difference of 71.75% from the benchmark for this hour. In contrast, Case 1 gets closer to the benchmark after 4 min. The average power variation for Case 1 finally reaches 49.77 kWh, reducing the difference to 34.84%. The reason is that the proposed strategy can identify adverse trends in regulation signals and make corresponding responses. The average power variation of Case 2 is 21.36 kWh, which is close to that of Case 3. It can be seen that without the exponential decay signal, the strategy will be difficult to characterize the trend of future regulation signals. On the whole, the average power variation brought by the proposed strategy is much closer to D_t . Since D_t is obtained by the day-ahead scheduling with the goal of BSS economy, the proposed strategy can get more profits. In addition, the solution time for a single optimization is only 0.1 s, meeting the response requirements in real-time operation.

Next, we assess the response ratio to regulation signals which is related to the regulation performance. In Fig. 15, the response ratios in the three cases between 12:00 and 13:00 are compared. Overall, the average response ratio in Case 1 is 0.79, which exceeds the lower bound

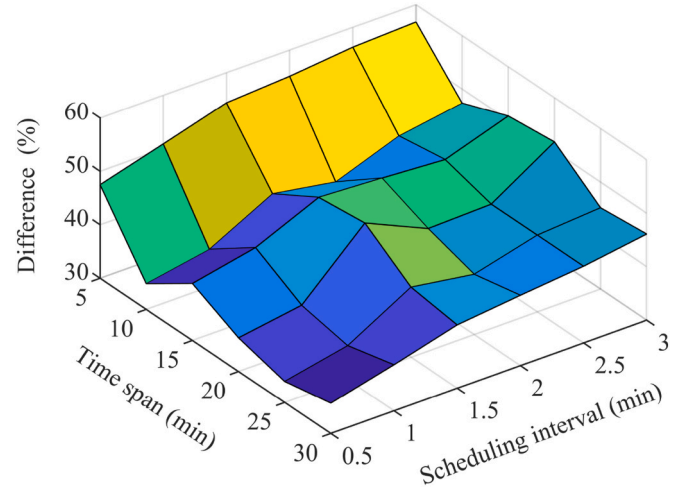


Fig. 16. Sensitivity analysis of real-time scheduling interval and time span for calculating γ_{τ}^{ave} .

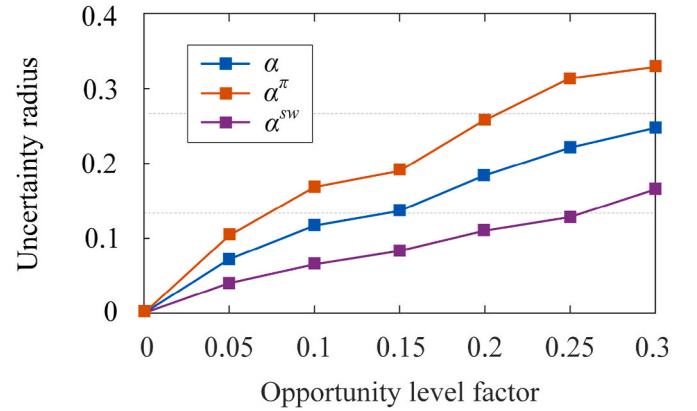


Fig. 17. Comparison of IGDT scheduling results under different opportunity level factors.

of 0.7. It can be observed that BSS fully responds to most positive regulation signals. However, due to D_t is a large positive number in that hour, the negative signals are partially responded to. The average response ratio in Case 2 is 0.74, which is 6.33% behind the regulation performance of Case 1. The gap mainly comes from the parts with high signal amplitude and dramatic changes. It indicates that the exponential decay signal can make the strategy more adaptable to signal fluctuations.

We compare the intra-day performances when the lower bound of the average response ratio γ_{τ}^{ave} takes different values. The results in Table 1 show that BSS revenue is inversely proportional to γ_{τ}^{ave} . When $\gamma_{\tau}^{ave} = 1$ (i.e., fully responding to the regulation signals), the average difference with the benchmark increases by 31.05% compared to $\gamma_{\tau}^{ave} = 0.6$, while the intra-day revenue decreases by \$79.18. It demonstrates that a stricter requirement on γ_{τ}^{ave} leads to better regulation performance at the cost of BSS's economic benefits. Operators can make a trade-off by adjusting the parameter.

Two factors must be determined in actual operation: how often to perform real-time scheduling and the time span for calculating the average response ratio. The difference between the actual power variation and the benchmark is still used to evaluate real-time scheduling performance. Fig. 16 shows the sensitivity analysis results for these two factors. Overall, shorter scheduling intervals are beneficial for improving performance. When the scheduling interval is reduced from 3 min to 30 s, the difference from the benchmark decreases by 12.67%.

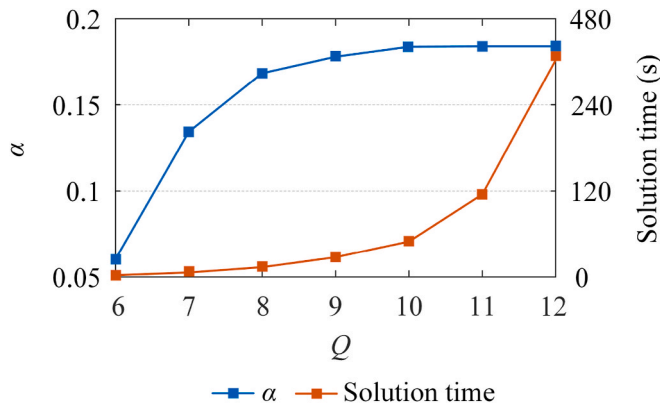


Fig. 18. The influence of Q variation on uncertainty radius and solution time.

Table 2

Comparison of the proposed method with stochastic optimization and robust optimization.

Methods	SP	RO	Proposed
Total reserve capacities (MWh)	11.34	10.86	12.32
Service revenue (\$)	400.62	393.15	402.3
Energy cost (\$)	330.87	340.4	324.48
Intra-day revenue (\$)	433.9	424.38	469.73
Solution time (s)	313.91	105.17	66.3

Increasing the time span generally improves performance, but the effect is less significant than reducing the scheduling interval. The best performance is achieved when real-time scheduling is performed every 30 s, and the average response ratio is calculated over the last 25 min.

6.4. Performance analysis of the proposed IGDT model

We first analyze two key parameters. In the IGDT model, the operator sets the opportunity level factor σ based on their risk-seeking preference. Fig. 17 compares the uncertainty radius α under different σ . It can be observed that α is directly proportional to σ . When BSS seeks more profit, the model searches for feasible solutions within a larger uncertainty set. Note that the uncertainty radius of HCRS is larger than that of the swapping demand. The reason is that the swapping demand must be an integer. When α^{sw} increases slightly, integer variables may not correspondingly change as HCRS does. Increasing the weight of α^x is advantageous for reducing sensitivity to HCRS. The fluctuation of local swapping demand needs to be considered when determining the weights.

The second key parameter is the number of binary variables Q in the quasi-precision method. As shown in Fig. 18, the objective function α changes significantly when Q increases from 6 to 9. Afterward, α converges to 0.1839. It suggests that the errors introduced by the binary representation can be negligible when $Q \geq 10$. Even when the reserve capacity is at its maximum value of 1200 kW, the error introduced by the quasi-precision method will not exceed 1.17 kW. However, the solution time shows a significant increase as Q grows. Overall, $Q = 10$ strikes a balance between accuracy and computational complexity.

Next, we compare the proposed IGDT method with stochastic programming (SP) and robust optimization (RO) to validate its capability in handling uncertainties. In SP, massive HCRS and swapping demand curves are reduced to several typical scenarios using classical K-means clustering. RO, on the other hand, searches for the worst scenario in an empirically given uncertain set. Results in Table 2 suggest that the proposed method outperforms SP and RO in various performances. The provided reserve capacities and intra-day revenue by SP are 0.98 MW and \$35.83 lower than the proposed method, respectively; for RO, it is 1.46 MW and \$45.35 lower, respectively. It proves that the typical

scenarios in SP cannot represent the randomness of HCRS; meanwhile, the worst scenario selected by RO makes the reserve capacity scheme too conservative. Additionally, the solution time of the proposed method is only 66.3 s, which is only 21.12% and 63.04% of SP and RO.

7. Conclusions

This paper investigated the day-ahead bidding and real-time response strategies for BSS participation in the frequency regulation market. A new uncertainty set is designed for dynamic regulation resources. An opportunity model for optimizing the reserve capacities is constructed based on IGDT and solved by the proposed BAS method effectively. Additionally, a lightweight real-time response strategy is proposed, which can maximize BSS profits while keeping high regulation performance. Simulation results validated the superiority of the reserve capacities obtained through the IGDT-based day-ahead scheduling method. It achieves higher daily revenue and significantly reduces solution time compared to the traditional stochastic optimization method. In real-time response, the proposed strategy gets an additional 16.46% revenue compared to the full response strategy, while the average response ratio is above 80%. The real-time strategy can be solved within 0.1 s, which ensures a fast response speed in practical applications. Furthermore, the strategies proposed in this study can also be applied to other dynamic regulation resources except BSS.

CRedit authorship contribution statement

Ziqi Wang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation. **Sizu Hou:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the National Key R&D Program of China (No. 2018YFF01011900).

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